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Assignment No: 02

Question 01: Is there any clear gradient within species and functional traits?

Answer: Yes. The trait ordination shows a clear first axis gradient (explaining 26.56% of variation) separating dispersal modes (anemochory vs zoochory) and the second axis (explaining 20.93% of variation) is driven by seed/fruit size/mass. So there is a clear functional gradient in the trait space.

Question 02: Which trait has higher association with axis 1? and which with axis 2?

Answer: This can be seen in the following table. The higher the value of a trait for a spcefic axis, the higherthe contribution to it.

	CS1 <dbl>	CS2 <dbl>
Shade.Tolerant	-0.08591572	0.4075974
Shade.Intolerant	0.22793781	-0.3525852
Shade.Intermediate	-0.10656440	-0.1385359
Zoochorous	-0.56485336	0.1050457
Anemochorous	0.56485336	-0.1050457
Seed.length	0.13913004	0.1106864
Seed.mass	-0.12582680	0.4529875
Fruit.length	0.40935155	0.2584914
Fruit.width	0.28596356	0.4096592
Fruit.mass	0.08157720	0.4649721

For example:

- **Axis 1:** Dispersal mode, specifically Zoochorous (strong negative loading) vs Anemochorous (strong positive loading) are the largest loadings on axis 1.
- **Axis 2:** Fruit / seed size & mass (e.g. Fruit.mass, Seed.mass, Fruit.width) show the largest associations with axis 2.

Question 03: Are anemochorous species more shade intolerant?

Answer: Yes, there is an association: anemochorous species project on the same side of axis 1 as Shade.Intolerant (both have positive loadings on axis 1), which indicates that anemochorous species tend to be more shade-intolerant in this dataset. (The loadings for Anemochorous and Shade.Intolerant on axis 1 are both positive).

Question 04: What is the main trend in this ordination?

Answer: The main trend reveals a forest fragmentation gradient explaining 70.56% of environmental variation. Sites arrange continuously from small, fragmented patches with high edge effects (Small sites, left side) to large, intact forests with interior conditions (Large sites, right side) (CS1), with a secondary gradient (CS2) reflecting forest structure from dense, tall forest to more open, shorter forest.

Question 05: Based on the ordination of site on trait space, can you describe the main gradient between site and traits?

Answer: The main gradient in community-weighted trait space (CWM) separates communities by dispersal mode and seed/fruit size i.e., sites differ mainly in whether they are dominated by wind-dispersed, shade-intolerant species or by animal-dispersed, larger-seed species. The environment explains a large fraction of this trait variation.

Question 06: How much variance is shown by PC1 explaining the variance of traits by sites parameters?

Answer: PC1 of the community-weighted PCA (pca.cwm) accounts for ≈46.57% of that trait variance.

Question 07: How would you explain the positive correlation between area of fragments and seeds/fruits?

Answer: Larger forest fragments tend to sustain more interior-forest conditions (shadier, more stable microclimate, more animal dispersers), which favors larger-seed / larger-fruit species (often animal-dispersed) because larger seeds have better shade-tolerant recruitment and competitive ability in closed canopy.

Question 08: Why seeds dispersed by wind are negatively correlated with the area of the forest?

Answer: wind-dispersed species are often early-successional, open-habitat or edge specialists that do better in smaller/fragmented or more open patches where wind dispersal is effective and light is abundant. Thus, as fragment area increases (more closed, interior conditions), the relative abundance of anemochorous species tends to decrease, reflected by the negative association (area vs anemochorous)